

Fernando Fernández Andueza

Gameplay & AI Programmer

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Valencia, Spain · EU citizen · open to relocation within Spain and to remote work · Spanish (native)

PROFILE

Gameplay and AI programmer. I built the enemy AI for Beerserker: The Stolen Brew (Early Access on Steam, team of about 20): all four archetypes, in StateTree with C++ and Blueprints. I built XEMA, my own C++ and OpenGL engine, with an ECS and deferred PBR rendering. I work with Unreal and Unity, and before games I spent two and a half years as a fullstack developer on ERP software.

TECHNICAL SKILLS

Engines & AI	Unreal Engine 5 (C++ & Blueprints), Unity (C#), XEMA — custom C++ engine, StateTree, Behaviour Trees, EQS, AI Perception, Animation systems, Object pooling
Languages	C++, C#, C, Lua
Graphics & systems	OpenGL 4.5 DSA, Deferred shading / PBR, Vertex & fragment shaders, ECS architecture, ImGui, GLFW, Multithreading & job systems, Memory management & pointers, PhysX 5
Tools & platforms	Unreal Insights, Visual Studio debugger, RenderDoc, PC (Windows), Nintendo Switch (emulator), PlayStation 4 (partial, devkit), Git, Perforce, Plastic SCM, Premake5
Outside games	TypeScript / JavaScript, HTML / CSS, Vue.js, SQL Server, Java, VB .NET

EXPERIENCE

Gameplay & AI Programmer · Beerserker: The Stolen Brew Sept 2025 – Jul 2026

- ESAT final-year project, team of about 20 · developed by Grumpy Games and ESAT, published on Steam by ESAT (Early Access, free)
- Implemented the four enemy archetypes (melee, ranged, explosive and a tank mini-boss) with StateTree in C++ and Blueprints, with custom tasks and conditions.
 - Built a templated object pool as a plugin, reusable in other projects, for the many enemies, particles, decals and projectiles.
 - Used EQS to pick where the ranged enemy retreats to, based on distance to the player and line of sight. The tank favours its less-used melee attacks, so it doesn't keep repeating the same one.
 - Built the enemy spawners, set up and uploaded the Steam builds, and took part in designing the architecture of a custom GAS-based action system.

Gameplay Programmer · Roadtrip: The Engine of Madness · Oniric Tales Dec 2024 – Jun 2025

- Designed and built core gameplay systems: a component-based dialogue system, an AI perception system integrated with the behaviour trees, and a radio communication system.
- Also worked on the Behavior Graph, general gameplay mechanics, profiling and bug fixing.

Fullstack Developer · AHORA Sept 2020 – Mar 2023

- Enterprise management system: frontend with Vue.js, TypeScript and DevExtreme; backend in VB and C# on MSSQL.
- Built products, components and extensions with FlexyGo, an in-house low-code framework.
- Handled client incidents, often on site. Worked in Scrum.

SELECTED PROJECTS

XEMA — custom 3D engine C++ · OpenGL 4.5 · Nintendo Switch (emulator)

- ECS, deferred PBR, shadow mapping, instancing, a JobSystem for async mesh loading, Lua scripting and PhysX.
- Ported to Nintendo Switch, tested only on an emulator. PS4 port in progress on a devkit: the ECS and simple-geometry rendering work; mesh loading, textures, instancing, lights, shadows and PBR are still missing.

Abandoned Hospital Unreal Engine 5 · C++ · AI Perception, EQS

- Class exercise with AI Perception and EQS on a behaviour tree; I later rebuilt it on my own, with the AI rewritten in StateTree.

EDUCATION & CERTIFICATIONS

HND in Video Game Programming · ESAT — Escuela Superior de Arte y Tecnología · Valencia 2023 – 2026

Video game programming, specialising in C++: my own graphics engine from scratch, procedural generation (Wave Function Collapse, cellular automata, L-systems), a templated FSM, A* pathfinding, and assembly to find and fix bottlenecks in inner loops.

DAM — Multiplatform Application Development · IES El Grao · Valencia 2018 – 2020

Java for desktop and Android, relational databases, HTML and CSS, OOP fundamentals and version control.

Certified Scrum Master & Product Owner · Scrum.org / International Agile Institutes Professional credentials